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Conceptualizing collectivity in lattice resonances of periodic arrays of nanostructures

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E-mail: andrea.toma@iit.it and a.manjavacas@csic.es**Keywords:** collectivity, lattice resonances, periodic arrays, nanoparticle arrays, plasmonic crystalsSupplementary material for this article is available [online](#)

Abstract

The use of periodic arrays of nanostructures has become a powerful strategy for enhancing light–matter interactions. These systems support collective modes known as lattice resonances, which are characterized by strong and spectrally narrow optical responses. While all lattice resonances arise from the multiple, coherent scattering among the nanostructures of the array, the particular geometrical characteristics of the system can strongly affect the interaction strength, leading to very distinct optical responses. Despite the extensive research on this topic, a consistent framework for quantifying and comparing these differences remains lacking. In this work, we introduce the degree of collectivity of a lattice resonance, defined as a measure of the interaction strength between the nanostructures in the array, and relate it to the spectral detuning of the lattice resonance—an easily quantifiable metric. This allows us to establish a direct link between the geometrical characteristics of the array and the optical properties of the lattice resonances it supports. Through a combination of theoretical modeling and experimental validation, we investigate how the degree of collectivity affects the robustness of lattice resonances when the system deviates from perfect periodicity. Our results show that lattice resonances with higher degree of collectivity are more sensitive to common experimental imperfections, such as finite array size, positional disorder, and refractive index mismatch between the substrate and the superstrate. The degree of collectivity introduced in this study provides a valuable tool for predicting the performance and experimental feasibility of the lattice resonances supported by periodic arrays of nanostructures, thus paving the way for the development of advanced photonic devices.

1. Introduction

When metallic nanostructures are arranged in a periodic fashion, they can support collective modes known as lattice resonances [1–7]. These modes stem from the collective, coherent interaction between the constituents of the array, and appear in the optical spectrum at wavelengths commensurate with the period of the array [5–8]. Furthermore, as a result of their collective nature, lattice resonances exhibit exceptionally strong field enhancements [9–12], and significantly narrower spectral features compared to the localized plasmon resonances of the individual nanostructures [13–16]. Owing to their unique optical properties, lattice resonances are being harnessed in a wide range of applications, including optical sensing [17–19], energy harvesting [20–22], energy transfer [23, 24], strong coupling [25–27], and the realization of Bose–Einstein condensation [28, 29]. They have also been utilized in color filtering

[30, 31], light-to-heat transducers [32, 33], chiral photonic elements [34–36], light-emitting devices [37–41], and flat optical lenses [42], among other examples.

It is well established that lattice resonances originate from the coherent interactions among the nanostructures that compose the periodic array. However, the strength of these collective interactions is highly sensitive to various system characteristics, including the array period, the size and morphology of the nanostructures, and the refractive index of the surrounding environment. Surprisingly, despite the extensive body of research on lattice resonances, there remains a lack of both qualitative and quantitative understanding of how these structural and material parameters influence the strength of the interactions and, consequently, how this impacts the optical response of the system.

To bridge this gap, we introduce the degree of collectivity, $\mathcal{C}$, as a measure of the strength of the coherent interactions that give rise to lattice resonances in periodic arrays of nanostructures. By analyzing a canonical example consisting of an infinite square array of spherical nanostructures embedded in a homogeneous environment, modeled within the dipolar approximation, we quantify the strength of the interaction required to sustain the corresponding lattice resonance. Then, we demonstrate that the spectral detuning between the lattice resonance and the corresponding Rayleigh anomaly, Δ_{LR} , serves as a reliable indicator of the degree of collectivity. Specifically, smaller spectral detunings correspond to higher degrees of collectivity, reflecting stronger coherent coupling among the nanostructures. This relationship positions $\mathcal{C}$ as a theoretical construct that validates the use of Δ_{LR} , a far more accessible metric that we leverage throughout the remainder of the work. Equipped with this understanding, we examine the impact of collectivity in arrays with a finite number of constituents. Our study reveals that lattice resonances with a higher degree of collectivity require a larger number of nanostructures to approximate the response of an infinite array. Furthermore, when introducing disorder in the position of the nanostructures, we find that more collective lattice resonances are more susceptible to such imperfections, as evidenced by a significant reduction in their quality factor. This prediction is experimentally verified by analyzing arrays of nanodisks with varying levels of disorder, thereby confirming the generality of our theoretical framework.

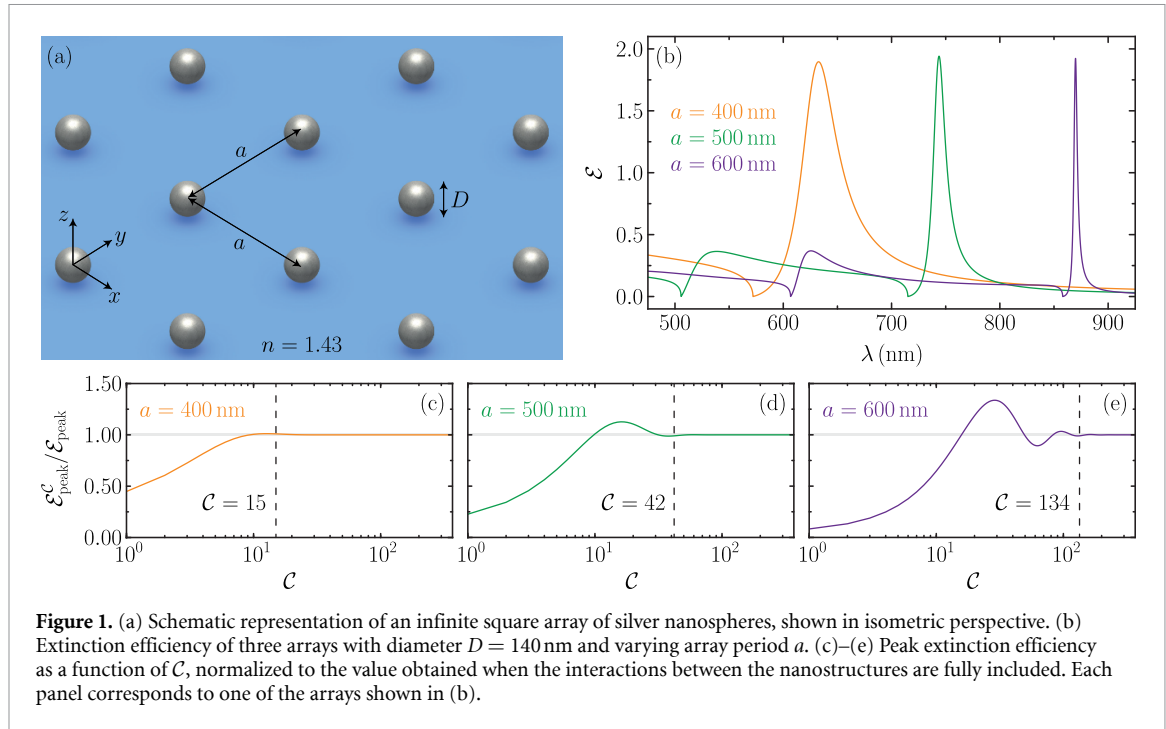
In addition, we examine the influence of refractive index mismatch between the substrate and superstrate on lattice resonances with varying degrees of collectivity. To this end, we compare the optical response of fabricated arrays of nanodisks with full-wave calculations of the corresponding systems. As expected, we find that lattice resonances with a higher degree of collectivity are more sensitive to refractive index mismatch, with their optical response exhibiting greater degradation as the mismatch increases. Furthermore, we propose that artificially increasing the refractive index mismatch in the full-wave calculations can effectively account for other experimental imperfections, such as finite-size effects, positional disorder, deviations from ideal plane-wave illumination [43], and other uncontrolled variations inherent to the fabricated system.

Altogether, the degree of collectivity introduced in this work provides a practical framework for assessing the feasibility of experiments involving lattice resonances. Our work advances previous studies addressing the effects of finite-sized arrays [44–50], positional disorder [47, 50–52], and refractive index mismatch between substrate and superstrate [53–56], by introducing a complementary perspective that connects the degree of collectivity, $\mathcal{C}$, a theoretically grounded metric, with the spectral detuning, Δ_{LR} , which can be readily obtained from numerically computed or experimentally measured spectra. This, in turn, enables a more accurate interpretation of empirical results and ultimately guides the design of robust, high-performance photonic devices.

2. Results and discussion

We begin by considering the canonical case of an infinite square array of silver nanospheres of diameter D , as depicted in figure 1(a). The array lies in the xy -plane, has a period a , and is embedded in a homogeneous medium with a refractive index of $n = 1.43$, representative of the experimental conditions analyzed later. For the dielectric function of silver, we use the tabulated data from reference [57]. The system is excited at normal incidence by a plane wave polarized along the x -axis. We assume that the size of the nanostructures is significantly smaller than both a and the wavelength of the incident field λ in vacuum. Within these conditions, the system can be accurately described using the well-established coupled dipole model (CDM) [24, 58–62], in which each nanostructure is modeled as a point electric dipole with polarizability α . Owing to the symmetry of the array and the excitation conditions, the dipole induced in each of the nanostructures is aligned along the x -axis and has the same value, given by

$$p = \mathcal{A}E, \quad (1)$$



where E is the incident electric field amplitude, and

$$\mathcal{A} = [\alpha^{-1} - \mathcal{G}]^{-1}, \quad (2)$$

represents the effective polarizability of the array [63]. This quantity captures the interplay between the individual response of the nanostructures, encoded in α , and their collective electromagnetic interaction, described by the lattice sum $\mathcal{G}$. The latter is defined as $\mathcal{G} = \sum_{i \neq 0} G(\mathbf{R}_i)$, where $\mathbf{R}_i$ represent the position of the nanostructures and $G(\mathbf{R}) = [k^2 + n^{-2} \partial_x \partial_x] \exp(ikn|\mathbf{R}|)/|\mathbf{R}|$, with $k = 2\pi/\lambda$, is the xx -component of the dipole–dipole interaction tensor. In this work, we compute the lattice sum $\mathcal{G}$ using Ewald’s summation method [64], which ensures high accuracy and rapid convergence. Note that we only need to consider the xx -component of $\mathcal{A}$, as the xy -components vanish due to the symmetry of the square array.

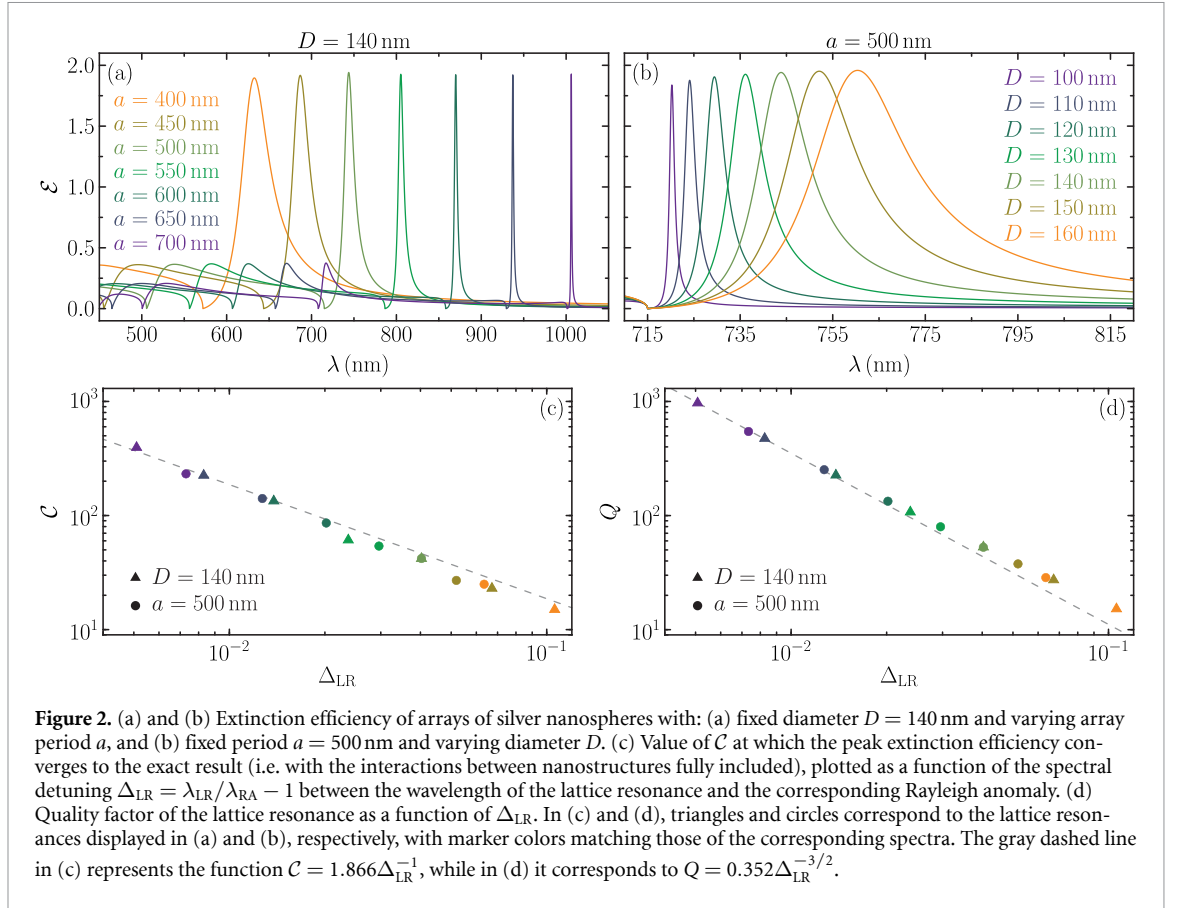
We can exploit the properties of the geometric series to transform equation (2) into

$$\mathcal{A}_\mathcal{C} = \sum_{s=0}^{\mathcal{C}} (\alpha \mathcal{G})^s \alpha. \quad (3)$$

This transformation is valid only within the frequency range where the condition $|\alpha \mathcal{G}| < 1$ is satisfied, ensuring the convergence of the geometric series.

Equation (3) allows us to quantify the role of the interactions between the nanostructures in the optical response of the array. Specifically, $\mathcal{C} = 0$ corresponds to the case in which interactions are entirely neglected (i.e. $p = \alpha E$), while $\mathcal{C} = \infty$ fully incorporates them (i.e. $p = \mathcal{A} E$). Therefore, equation (3) implicitly defines the level of collectivity as the smallest integer $\mathcal{C}$ for which the resulting response function reproduces, within a specified tolerance, the behavior of the array that fully incorporates the interactions among the nanostructures.

Figure 1(b) shows the extinction spectra of three arrays with $D = 140$ nm and periods $a = 400$ nm (orange curve), $a = 500$ nm (green curve), and $a = 600$ nm (purple curve), calculated using the CDM as $\mathcal{E} = \text{Im}\{p\}/\Pi$ with $\Pi = a^2 \lambda n E / (8\pi^2)$. The extinction efficiency represents the ratio of the power extracted from the incident field, through both scattering and absorption, to the incident power. As shown in figure S2 of the supplementary material, these results are in excellent agreement with full-wave numerical solutions of Maxwell’s equations obtained using the finite element method (FEM). Examining these spectra, we observe that increasing the array period a while keeping the nanostructure diameter D fixed results in a lattice resonance with a progressively narrower lineshape, whose extinction consistently reaches a value close to the theoretical maximum of 2 [62]. For each of these lattice resonances, figures 1(c)–(e) show the evolution of the peak extinction efficiency as a function of $\mathcal{C}$, calculated using the effective polarizability $\mathcal{A}_\mathcal{C}$ defined in equation (3). The results are normalized to the corresponding values obtained using $\mathcal{A}$ (i.e. for $\mathcal{C} = \infty$), which fully accounts for the interactions between the



nanostructures. We focus on the peak extinction efficiency because, in all cases considered, the peak of the lattice resonance falls within the spectral range where the geometric series expansion in equation (3) remains valid. As shown in figure S3 of the supplementary material, the condition $|\alpha\mathcal{G}| < 1$, which ensures the convergence of the expansion, is not satisfied across the entire spectrum. This makes it impractical to compute full spectra as a function of C .

The peak extinction efficiency for the array with the smallest period converges to the exact result at $C = 15$. We define convergence as the value of C beyond which the peak extinction efficiency deviates by less than 1% from the result obtained with full interaction. Larger values of C are required to achieve convergence for arrays with larger periods, specifically $C = 42$ for $a = 500$ nm and $C = 134$ for $a = 600$ nm.

Therefore, we infer that, for a fixed diameter, the degree of collectivity of the lattice resonance increases with the array period, and is accompanied by a narrowing of the lineshape. This is confirmed by the extinction efficiency spectra displayed in figure 2(a) for seven arrays with $D = 140$ nm and periods comprised between 400 nm and 700 nm. Moreover, as a increases, the spectral position of the lattice resonance shifts closer to the first Rayleigh anomaly of the array, which corresponds to the wavelength at which the first diffraction order becomes grazing, given by $\lambda_{RA} = an$. A similar trend is observed when the period is held constant and the nanostructure diameter is decreased, as shown in figure 2(b) for $a = 500$ nm, in which smaller particles lead to narrower resonances that approach the corresponding Rayleigh anomaly located at $\lambda_{RA} = 715$ nm.

Indeed, figure 2(c) reveals a clear functional relationship between the value of C at which convergence is achieved and the spectral detuning $\Delta_{LR} = \lambda_{LR}/\lambda_{RA} - 1$, which quantifies the separation between the wavelength of the lattice resonance λ_{LR} , and the corresponding Rayleigh anomaly. In this figure, triangles and circles represent the lattice resonances from figures 2(a) and (b), respectively, with marker colors matching those of the corresponding spectra. From these results, we infer that C is inversely proportional to Δ_{LR} , a trend confirmed by the best-fit curve shown as a dashed gray line, corresponding to the relation $C = 1.866\Delta_{LR}^{-1}$.

These results demonstrate that Δ_{LR} serves as a reliable predictor of the degree of collectivity of a lattice resonance. This is particularly relevant since C is defined within the CDM and is therefore restricted to systems for which this approximation accurately captures the optical response. Although, in principle,

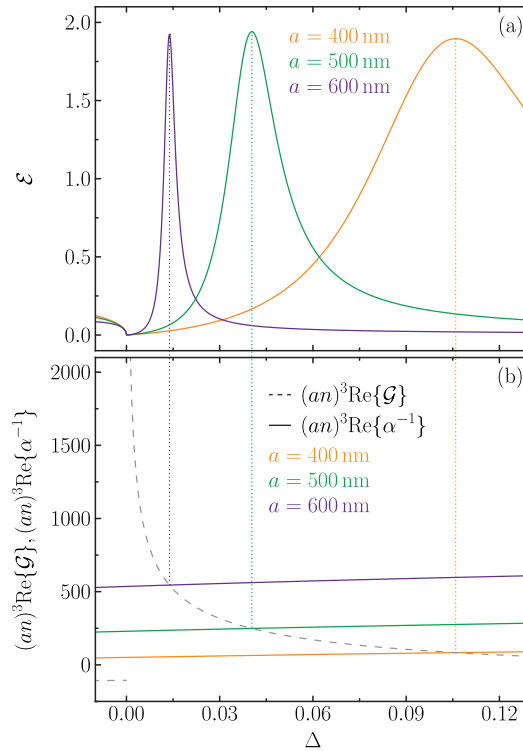


Figure 3. (a) Extinction efficiency of three arrays of silver nanospheres with $D = 140$ nm and different periods, as indicated in the legend. (b) Real parts of the lattice sum $\text{Re}\{\mathcal{G}\}$ (dashed gray curve) and the inverse polarizability $\text{Re}\{\alpha^{-1}\}$ (solid colored curves) for the arrays of (a). Both quantities are plotted as a function of $\Delta = \lambda/\lambda_{\text{RA}} - 1$ and scaled by $(an)^3$ to facilitate the comparison. The vertical dotted lines indicate the spectral positions where $\text{Re}\{\alpha^{-1} - \mathcal{G}\} = 0$, which approximately determine the wavelengths of the respective lattice resonances.

$\mathcal{C}$ can be generalized to more sophisticated methods such as those based on the T -matrix formalism [65], which account for higher-order multipoles, and can be adapted to arrays composed of nanostructures of different materials and geometries by using the appropriate polarizability, it nonetheless remains tied to a specific modeling framework. In contrast, Δ_{LR} can be directly computed for any array using full-wave numerical solutions of Maxwell's equations. This is demonstrated later in this work through a FEM simulation of an array of nanodisks. Moreover, since both the lattice-resonance wavelength and the Rayleigh anomaly can be identified from commonly measured experimental spectra, such as reflectance and transmittance, Δ_{LR} can also be directly measured, making it a practical and broadly applicable metric.

An important observation from figures 2(a)–(c) is that as the degree of collectivity increases, and consequently Δ_{LR} decreases, the lattice resonance exhibits a progressively narrower lineshape. This trend is clearly reflected in the quality factor, defined as the ratio of the resonance wavelength to its linewidth, as shown in figure 2(d). Consistent with our previous findings [11, 66], the quality factor approximately follows the scaling law $Q \sim \Delta_{\text{LR}}^{-3/2}$. This functional dependence is overlaid in the figure as a gray dashed curve, fitted to the data with a best-fit prefactor of 0.352. Note that this prefactor, as well as the previous one, depends on the material and morphology of the nanostructures, in addition to the geometry and environment of the array. As a result, quantitative comparisons of Δ_{LR} , $\mathcal{C}$, and Q are only meaningful between arrays where the aforementioned parameters are kept constant, thus serving as relative measures rather than absolute, system-independent metrics.

We can understand the relationship between Δ_{LR} and the degree of collectivity by analyzing the physical origin of lattice resonances. These modes appear at the poles of the effective polarizability of the array $\mathcal{A} = [\alpha^{-1} - \mathcal{G}]^{-1}$ [63]. As we mentioned before, this quantity depends on both the individual response of the nanostructures, characterized by their polarizability α , and the collective interactions captured by the lattice sum $\mathcal{G}$. Consequently, the value of $\mathcal{G}$ at the pole associated with a lattice resonance determines the strength of the interaction between the nanostructures and, hence, the degree of collectivity of the resonance. Under the illumination conditions considered in this study, namely, normal incidence and linear polarization, the lattice resonance appears approximately at the wavelength for which $\text{Re}\{\alpha^{-1} - \mathcal{G}\} = 0$. This is illustrated in figure 3. Specifically, figure 3(a) presents

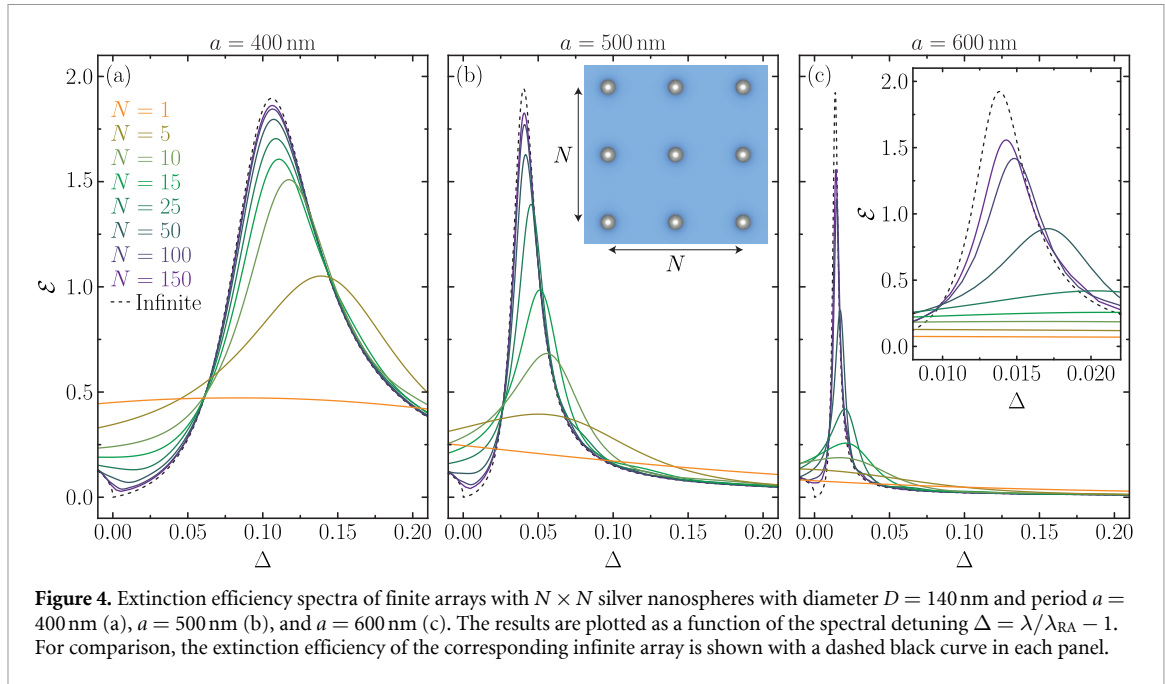


Figure 4. Extinction efficiency spectra of finite arrays with $N \times N$ silver nanospheres with diameter $D = 140$ nm and period $a = 400$ nm (a), $a = 500$ nm (b), and $a = 600$ nm (c). The results are plotted as a function of the spectral detuning $\Delta = \lambda/\lambda_{\text{RA}} - 1$. For comparison, the extinction efficiency of the corresponding infinite array is shown with a dashed black curve in each panel.

the extinction efficiency of three arrays with $D = 140$ nm and varying array periods, while figure 3(b) shows the corresponding values of $\text{Re}\{\mathcal{G}\}$ and $\text{Re}\{\alpha^{-1}\}$ scaled by $(an)^3$ to facilitate direct comparison. As anticipated, the condition $\text{Re}\{\alpha^{-1} - \mathcal{G}\} = 0$, marked by the vertical dotted lines, accurately signals the spectral position of the corresponding lattice resonance.

Figure 3(b) shows that the real part of the lattice sum rapidly increases as the wavelength approaches the Rayleigh anomaly, where its value diverges. Therefore, a lattice resonance appearing closer to the Rayleigh anomaly originates from a stronger collective interaction between the constituents of the array, thus exhibiting a higher degree of collectivity. Moreover, the slope of $\text{Re}\{\mathcal{G}\}$ also increases as the wavelength approaches the Rayleigh anomaly. This implies that the wavelength range over which the condition $\text{Re}\{\alpha^{-1} - \mathcal{G}\} = 0$ is approximately satisfied becomes narrower, resulting in a sharper resonance and, hence, a higher quality factor. These results also indicate that, since for highly collective lattice resonances, $\text{Re}\{\mathcal{G}\}$ varies much more rapidly with wavelength than $\text{Re}\{\alpha^{-1}\}$, positional disorder is expected to have a more significant impact than disorder in the size of the nanostructures.

Experimental realizations of periodic arrays inevitably depart from the idealized case of an infinite, perfectly-periodic system. Finite array size, positional disorder of the nanostructures, and refractive index mismatch between the superstrate and the substrate all introduce variations in the optical response of the system. These factors, though distinct, share a common consequence: they weaken the collective interaction between nanostructures and thus reduce the strength of the lattice sum. As a result, the lattice resonances of different arrays may respond in very distinct ways to such perturbations, thereby motivating an examination of how the degree of collectivity dictates the robustness of lattice resonances under realistic experimental conditions. Through this analysis, we aim to establish a reliable framework that facilitates meaningful comparisons with empirical measurements.

We first consider finite arrays with $N \times N$ silver nanospheres embedded in a homogeneous medium with a refractive index of $n = 1.43$. We analyze the extinction efficiency, defined as $\mathcal{E} = \text{Im}\{p\}/\Pi$, where p is the total dipole moment of the array for finite systems, while for infinite systems is given by equation (1), as explained above. Unlike transmittance or reflectance, the extinction efficiency is well defined for both finite and infinite arrays, thereby enabling a consistent and meaningful comparison between them [43, 47]. Figures 4(a)–(c) show $\mathcal{E}$ for three arrays with $D = 140$ nm and periods $a = 400$ nm, 500 nm, and 600 nm, respectively. In each case, we analyze a range of sizes, from a single nanosphere to arrays with up to 150×150 elements, and include the infinite-array spectrum for reference.

Based on the analysis in figure 1, the system with $a = 400$ nm supports the lattice resonance with the lowest degree of collectivity, as confirmed by its largest spectral detuning from the Rayleigh anomaly. As shown in figure 4(a), the array with 10×10 nanospheres already reaches 80% of the peak extinction efficiency of the infinite system, and this value increases to 95% for the 50×50 array. Alongside the increase in peak extinction efficiency, the resonance linewidth narrows, approaching that of the infinite

array. Notably, the resonance position progressively blueshifts toward the Rayleigh anomaly, consistent with the strengthening of collective interactions as the number of elements in the array increases.

When considering a more collective lattice resonance, such as that of the system with $a = 500$ nm shown in figure 4(b), the dependence on the number of elements becomes markedly stronger. Arrays with fewer elements, which previously exhibited a response close to that of the infinite system, now show substantial deviations. For example, the peak extinction efficiency of the array with 50×50 elements achieves 84% of the infinite case, and only when N increases to 100, this number surpasses 90%.

For the lattice resonance with the highest degree of collectivity under consideration, which corresponds to the system with $a = 600$ nm and is analyzed in figure 4(c), these effects become even more pronounced. In particular, the spectra of arrays with $N < 10$ show no evidence of a lattice resonance. The array with 50×50 elements achieves only 46% of the peak extinction efficiency of the infinite system, while the 100×100 and 150×150 arrays reach 74% and 81%, respectively (see inset).

Based on these results, we conclude that the higher the degree of collectivity of a lattice resonance, the larger the number of particles required in the array to reproduce the response of the infinite system. This behavior is consistent with the fact that the degree of collectivity is linked to the strength of the interactions between nanostructures. A higher degree of collectivity necessarily implies that more nanostructures must interact coherently. From an experimental standpoint, this result imposes an important constraint. Since theoretical designs are typically based on infinite, perfectly-periodic arrays, there is a minimum number of elements required to sustain a lattice resonance, which depends on its degree of collectivity. This threshold must be met in order to achieve a response that closely approximates the predictions of theoretical calculations.

Beyond finite array size, the positional disorder of the nanostructures in the array drives further departures of the optical response of the lattice resonance with respect to the infinite, perfectly-periodic system. Since a nonuniform spacing between nanostructures hinders the extent over which the collective coupling can take place, we anticipate that more collective resonances, those requiring the coherent interaction of many elements, are more sensitive to positional disorder. To investigate this effect, we model positional disorder by first generating an ordered finite array and then introducing random displacements to each nanostructure along the x - and y -axes relative to its nominal position. These displacements are drawn from a Gaussian distribution such that the distance between neighboring nanostructures has a mean of a and a standard deviation of βa , where β is a parameter that quantifies the level of disorder.

Figure 5(a) presents the quality factor of the lattice resonance supported by arrays with different periods as a function of the disorder parameter β . These results are obtained from the calculation of the optical response of finite arrays consisting of 100×100 silver nanospheres with diameter $D = 140$ nm, embedded in a homogeneous medium with refractive index $n = 1.43$. We consider five different array periods, as indicated in the legend. For each combination of a and β , we perform five independent calculations of the extinction efficiency of the system, each corresponding to a randomly generated distribution of nanostructures within the array. The resulting spectra are provided in figure S4 of the supplementary material. From each calculation, we extract the quality factor and calculate its average, which is indicated in figure 5(a) with a circle. As expected, increasing the level of disorder in a given configuration leads to a systematic reduction in the quality factor, which progressively approaches the value of Q for a single nanostructure, indicated by the gray dashed line. Consistent with our initial hypothesis, arrays supporting more collective lattice resonances are significantly more susceptible to positional disorder. This is evidenced by the increase of the slope with the array period. While for the array with $a = 400$ nm, the quality factor decreases from approximately 15 at $\beta = 0$ to about 4 at $\beta = 0.2$, for the most collective lattice resonance examined, corresponding to the array with $a = 600$ nm, Q drops from around 194 to 49 over the same range of β .

Figure 5(b) shows a similar study for fabricated arrays. Specifically, we consider $100 \times 100 \mu\text{m}^2$ arrays of silver nanodisks with diameter $D = 140$ nm and height $h = 50$ nm. The arrays are fabricated on a CaF_2 substrate and encapsulated in a polydimethylsiloxane (PDMS) coating. As in the theoretical counterpart, the positions of the nanodisks are chosen by modeling the distance between neighboring nanostructures as a random variable that follows a Gaussian distribution with mean a and standard deviation βa . Figure 5(c) displays a few examples of the fabricated arrays. In this case, we extract the quality factor of the lattice resonance from the measured transmittance $\mathcal{T}$ (see figure S5 of the supplementary material for the corresponding spectra). Examining the results, we observe that, while the actual values of Q differ from those of the systems calculated in figure 5(a), owing to the differences in the morphology of the nanostructures and the dielectric environment, the overall trend remains consistent. Namely, arrays supporting more collective lattice resonances exhibit a more pronounced degradation in Q with increasing positional disorder. Notably, for certain configurations, the quality factor remains nearly unchanged

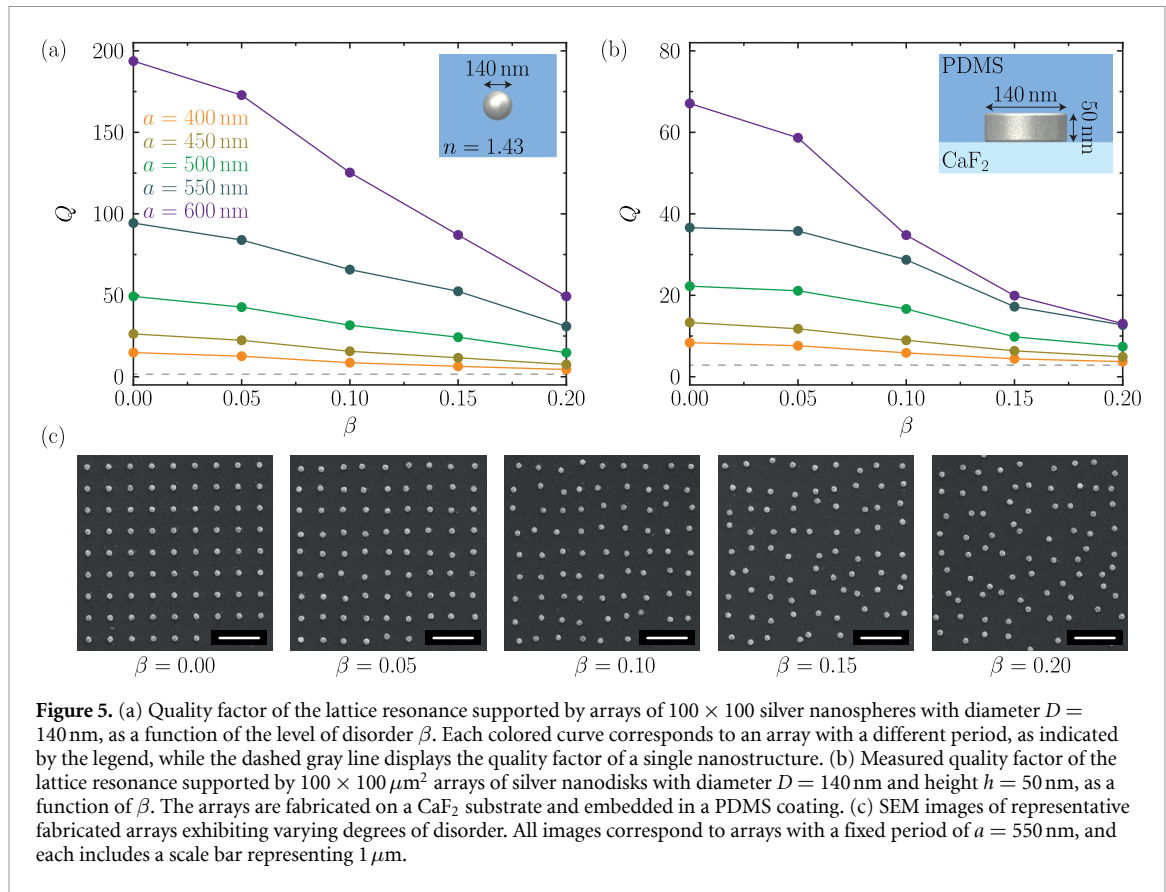


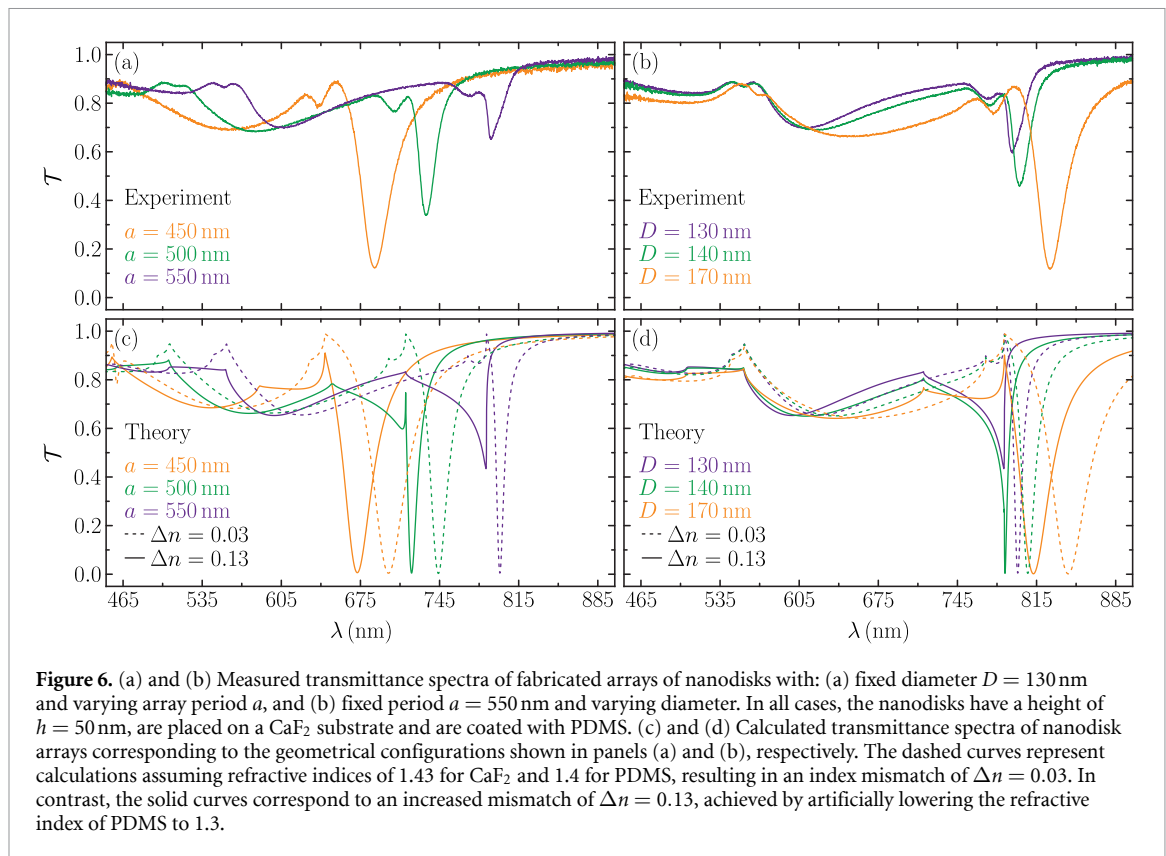
Figure 5. (a) Quality factor of the lattice resonance supported by arrays of 100 $\times$ 100 silver nanospheres with diameter $D = 140$ nm, as a function of the level of disorder β . Each colored curve corresponds to an array with a different period, as indicated by the legend, while the dashed gray line displays the quality factor of a single nanostructure. (b) Measured quality factor of the lattice resonance supported by 100 $\times$ 100 μm^2 arrays of silver nanodisks with diameter $D = 140$ nm and height $h = 50$ nm, as a function of β . The arrays are fabricated on a CaF_2 substrate and embedded in a PDMS coating. (c) SEM images of representative fabricated arrays exhibiting varying degrees of disorder. All images correspond to arrays with a fixed period of $a = 550$ nm, and each includes a scale bar representing 1 μm .

between $\beta = 0$ and $\beta = 0.05$, suggesting that the overall precision in the experimental setup is on the order of $\beta \sim 0.05$.

In any experimental realization, a periodic array of nanostructures must be supported by a substrate and surrounded by a superstrate, which rarely display identical refractive indices. While substrates are typically rigid solids, chosen for stability and compatibility with fabrication processes, superstrates are often air or liquids selected for optical accessibility or sensing applications, making refractive index alignment inherently difficult. This asymmetry in the surrounding dielectric environment introduces a refractive index mismatch that weakens the collective interaction between the nanostructures and, hence, degrades the properties of the lattice resonance [55]. Consequently, much like finite-size effects and positional disorder, the impact of refractive index mismatch is anticipated to vary significantly with the level of collectivity of the array.

Figures 6(a) and (b) display the measured transmittance spectra of arrays of nanodisks with period a , diameter D , and height $h = 50$ nm, fabricated on a CaF_2 substrate and coated with PDMS. In figure 6(a), the period a is varied while keeping $D = 130$ nm fixed. We observe that, as the period increases, the lattice resonance becomes increasingly degraded, as evidenced by the diminishing depth of the transmittance dip. This is consistent with the increasing degree of collectivity of the lattice resonance with a . A similar trend is observed in figure 6(b), where $a = 550$ nm is held constant and D is varied. Once again, the more collective lattice resonances, which in this case correspond to smaller values of D , for which the resonance peaks lie closer to the Rayleigh anomaly, exhibit the most pronounced degradation.

To complete our analysis, figures 6(c) and (d) present the calculated transmittance spectra corresponding to the same geometrical parameters as those used in the fabricated systems shown in figures 6(a) and (b). These results are obtained from full-wave solutions of Maxwell's equations, calculated using the FEM, which enables accurate modeling of the optical response of the array beyond the dipolar approximation, while accounting for the influence of both the substrate and superstrate. Unlike the CDM, however, FEM calculations are significantly more demanding in terms of computational resources and time, rendering the direct modeling of finite-sized arrays with hundreds of elements impractical. For this reason, in the FEM calculations, the array is modeled as infinite by defining a single unit cell and imposing periodic boundary conditions (see section S1 of the supplementary material for details). In figures 6(c) and (d), the dashed curves correspond to simulations using refractive indices of



1.43 for CaF_2 and 1.4 for PDMS, as taken from tabulated data [67, 68], resulting in an index mismatch of $\Delta n = 0.03$. Comparison with the experimental spectra reveals that this refractive index contrast does not fully reproduce the measured features. Improved agreement is achieved by increasing the mismatch to $\Delta n = 0.13$, implemented by artificially lowering the refractive index of PDMS to 1.3 in the calculations (see figure S6 of the supplementary material for results corresponding to other values of Δn). This adjustment effectively accounts for various experimental factors not explicitly included in the model, such as finite array size, structural disorder, limited illumination area, and other uncontrolled variations. As such, this approach offers a practical and efficient strategy for incorporating realistic experimental conditions while maintaining the idealized assumption of an infinite periodic system.

3. Conclusions

In summary, we have introduced the degree of collectivity for the lattice resonances supported by periodic arrays of nanostructures and investigated how this factor governs the robustness of these modes under typical experimental conditions, namely, finite array size, positional disorder, and refractive index mismatch between the substrate and superstrate. We have established a clear correlation between the degree of collectivity and the spectral detuning from the Rayleigh anomaly, with more collective modes appearing closer to the Rayleigh anomaly and exhibiting narrower linewidths. This relationship provides a reliable and accessible predictor of the degree of collectivity, which can be readily extracted from either experimental spectra or numerical calculations. Using combined theoretical modeling and experimental validation, we have demonstrated that lattice resonances with a higher degree of collectivity are more sensitive to imperfections and thus require larger and more precisely fabricated arrays for reliable observation, making their experimental realization more demanding. We have also found that while positional disorder degrades the lattice resonance of any array, systems supporting highly collective modes are particularly sensitive to such imperfections, for which even modest levels of disorder lead to a significant reduction in the quality factor. Furthermore, we have shown that refractive index mismatch between substrate and superstrate causes rapid degradation of highly collective lattice resonances, as evidenced by comparing the responses of fabricated nanodisk arrays and corresponding FEM simulations of infinite, perfectly-periodic arrays using identical geometrical and material parameters. Notably,

we have found that artificially increasing the refractive index mismatch in the calculations yields better agreement with experimental data. We interpret this adjustment as an effective way to account for experimental factors that cannot be explicitly included in the model due to the use of periodic boundary conditions, such as finite array size and structural disorder. By incorporating this effective parameter, we retain the computational efficiency of periodic simulations while achieving good agreement with experimental observations.

Overall, our work establishes a comprehensive and versatile framework for understanding the optical response of periodic nanostructure arrays through the lens of collectivity. By introducing a quantifiable measure of the degree of collectivity, in the form of the spectral detuning, we provide a powerful tool for predicting the experimental viability of theoretically designed lattice resonances. This framework enables researchers to assess, in advance, how sensitive a given resonance will be to real-world imperfections, which are often difficult to control in fabrication and measurement. Although our experiments use electron beam lithography, the insights from this work are broadly applicable to other fabrication methods [69], such as optical imprinting [70], self-assembly [52, 71], and soft nanoimprint lithography [72], where certain imperfections like structural disorder are more prevalent. By enabling a more accurate interpretation of experimental observations and clarifying the origins of discrepancies between theory and measurement, our approach provides a systematic framework for refining both modeling and design methodologies. The capability to anticipate and mitigate such deviations is essential for the dependable realization of photonic platforms that rely on the unique optical behavior of lattice resonances. Specifically, the spectral detuning can be incorporated into optimization algorithms as a constraint that assesses the feasibility of a proposed design, thereby preventing the algorithm from suggesting configurations that might appear optimal in theory but experimentally unattainable. Consequently, our findings deliver not only fundamental insights into the collective response of nanostructured arrays but also clear guidelines for the nanophotonics community, supporting the development of advanced photonic technologies for applications in sensing, light manipulation, and optical communications.

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Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

Conflict of interest

Authors state no conflicts of interest.

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